DSA 210 Introduction to Data Science
Project Proposal

Project Title: Predicting IMDb Movie Ratings Using Film Features

Motivation

The film industry generates enormous amounts of data, but predicting whether a film will be liked by audiences remains a challenging task. IMDb ratings are one of the most widely used indicators of movie quality and audience satisfaction. Understanding what drives these ratings can be valuable for producers, distributors, and streaming platforms. In this project, my goal is to explore objective film features such as budget, genre, runtime, and audience engagement metrics (vote count) can be used to accurately predict a movie's IMDb rating. This problem is both practically interesting and technically approachable, making it a feasible term project for applying machine learning techniques learned in this course.

Data Source

I will use two publicly available datasets for this project:

- **IMDb Dataset (from Kaggle / IMDb official data interface):** Contains movie titles, genres, ratings, number of votes, and runtime for thousands of films.
- **TMDB (The Movie Database) Dataset (from Kaggle):** Contains production budget, revenue, release date, and genre information. This dataset will be used to enrich the IMDb data with financial and production details.

Both datasets are publicly available on Kaggle and do not require any special access permissions. I will merge the two datasets using movie titles and release years as keys. The merged dataset is expected to contain approximately 5,000–10,000 movies with sufficient feature.

Planned Approach

The project will begin with downloading both the IMDb and TMDB datasets and merging them on movie title and release year, handling any duplicates that appear. After the merging process, missing values (especially missing budget entries) will be fixed and outliers will be removed to ensure data quality. Next, an exploratory data analysis phase will be conducted to visualize rating distributions, examine correlations between traits, and compare ratings across species. Multiple regression models—including Linear Regression, Random Forest, and Gradient Boosting—will be trained and evaluated using RMSE and R^2 metrics. Finally, the importance of the features will be analyzed to interpret which factors have the greatest impact on IMDb ratings.

Expected Outcome

My expectation is to create a model that can predict IMDb ratings with reasonable accuracy using selected features. I predict that the number of votes will show a strong correlation with rating (as popular films tend to have more stable scores), while genre and budget may reveal interesting but more variable relationships. The final deliverable will include a GitHub repository with all code, a Jupyter notebook with full analysis, and a written report.